

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

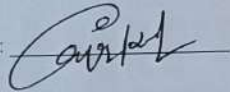
Total Marks: 25

Code of Conduct

I Y. Vinodhkumar (192422359) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

① Evolution from stand alone computing to cloud Based Systems.

cloud computing is the delivery of computing services over the Internet. It evolved from standalone computers through several technological developments

Evolution of computing:

- standalone computing
- client-server computing
- Internet computing
- virtualization
- cloud computing

Key technological transitions

- High speed Internet
- virtualization
- Data centers
- Automation
- APIs and web services

Advantages of cloud computing?

- cost effective
- Scalable
- Accessible from anywhere
- Reliable
- Easy maintenance

② Interpret the working of hypervisors in virtualization differentiate between type 1 and type 2 hypervisors with Examples

A hypervisor is software that creates and manages virtual machines (VMs). It allows multiple operating systems to run on one physical computer by sharing hardware resources such as memory storage and network.

The Hypervisors create isolated virtual environments that each VM works independently. if one VM crashes the other continue to work normally.

Type 1 Hypervisor (Base-metal):

- Installed directly on physical hardware
- Does not require a host operating system
- offers high performance and better security
- mainly used in cloud data centers and enterprises

Ex: VMware ESX:
Microsoft Hyper-V

Type-2 Hypervisor (Hosted):

- Installed on top of an existing operating system
- easier to install and use.
- suitable for personal computers, testing and learning
- slightly slower because it depends on the host OS

Examples: Oracle Virtual Box
VMware Workstation
Parallels Desktop

③ The role of API's in cloud service integration.

An application programming Interface (API) is a set of rules that allows different software applications to communicate with each other. In cloud computing APIs help different cloud services exchange information and work together.

Role of API's:

- connect different cloud services
- Enable communication b/w applications
- Automate cloud resource management
- share data securely
- support mobile and web applications

Interoperability: It means different systems can work together without compatibility problems

- using common standards such as REST and JSON.

1) The concept of resource pooling in cloud computing illustrate how it benefits multiple users.

Resource pooling is one of the important characteristic of cloud computing. It means that computing resources such as servers, storage, memory and networks are shared among multiple users using virtualization technology.

cloud providers maintain a pool of resources and allocate them to users whenever needed. users do not know the exact physical location of the resources.

working: Resources are allocated into a common pool.

- users request resources through the Internet
- The cloud automatically allocates resources
- when resources are no longer needed, they are returned to the pool

Benefits:

- Better utilization of hardware
- Lower operational cost
- High availability.

⑤ Interpret the concept of rapid elasticity in cloud computing.

Rapid elasticity is a key feature of cloud computing that allows computing resources to be increased or decreased automatically based on demand

working:

- users request cloud services
- The cloud monitors work load
- Resources are automatically increased during high demand
- Resources are reduced when demand because low

Benefits:

- Automatic scaling
- Better performance
- cost efficiency
- High availability
- No service interruption
- faster response to changing work to ads.